

Measuring Validator and Sandboxed-Execution Coverage for LLM-Generated NetOps Artifacts: A Paired Confirmatory Study

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Abstract—We preregistered and executed a paired confirmatory experiment (study_id f2c3d8b1-7a6e-4a9a-b2c3-5f8e1d2b6c7e) that measured the marginal detection benefit of adding sandboxed emulation to a deterministic validator for LLM-generated intent→configuration artifacts. Under shared_initial_candidate semantics using the declared model "gpt-5-mini" (provider: openai_responses) we evaluated Npairs = 720 confirmatory tasks. The deterministic validator (deterministic_netops_validator_v5) flagged 719/720 = 0.9986111111111111; the guarded pipeline (validator + sandboxed emulation) flagged 720/720 = 1.0. Paired difference (guarded - baseline) = 0.001388888888888884; exact McNemar two-sided p = 1.0; 95% bootstrap percentile CI (10000 resamples, seed = 1729) = [0.0, 0.004166666666666667]. Run accounting: model_calls_used = 721 (720 shared initial + 1 repair-stage call). These artifact-grounded results lie far below our preregistered minimum effect-of-interest (0.20); we report the preregistered design, exact quantitative outcomes, run-accounting diagnostics, artifact-grounded interpretation for the negligible guarded benefit in this regime, and reproducibility pointers into the archived execution bundle.

I. INTRODUCTION

Generative large language models (LLMs) are increasingly proposed to translate operator intent into device configuration and NetOps workflows. Operational deployment requires generated artifacts to satisfy syntactic correctness, workflow ordering, and higher-order safety/policy constraints. A common mitigation architecture is generate→validate→(repair)→execute, sometimes augmented by sandboxed emulation to reveal runtime-only failures that static validators miss. Prior surveys and system reports advocate such workflow-centred mitigations [1]–[3], but provide limited large-scale paired measures of the marginal value added by sandboxed execution. Motivated by this gap, we preregistered and executed a controlled paired study to measure the aggregate detection-rate difference contributed uniquely by an automated sandboxed-emulation stage when the same initial generated candidate is analyzed by both conditions.

Research question and contributions. Our preregistered research question was: for synthetic intent→configuration tasks under shared_initial_candidate semantics, what is the paired detection-rate difference (guarded - baseline) between (A) a deterministic validator-only pipeline and (B) the same deterministic validator followed by automated sandboxed emulation? Contributions: (1) a machine-readable preregistration and faithful autonomous execution; (2) an artifact-grounded

paired analysis on Npairs = 720 with complete accounting; (3) an artifact-grounded interpretation explaining a near-zero marginal effect; (4) concise reproducibility pointers into the archived bundle and prescriptive boundary conditions where guarded stages are likely to matter.

II. RELATED WORK

Workflow-centred mitigations and verification tools. Recent survey and architectural reports document the practical trend of combining LLM generation with deterministic validators, emulators, and repair loops to raise operational assurance for NetOps artifacts [1]. These works emphasize that correctness in intent→configuration pipelines is multidimensional: syntactic parsing, command-ordering/workflow constraints, invariants (e.g., keep-one-uplink-up), and dynamic runtime properties may each require different V&V primitives. Tool-focused case studies highlight the evolution and practical limits of static verifiers: mature configuration-analysis systems have progressively expanded rule-sets and topology-aware checks but still exhibit gaps for runtime-dependent behaviors and stateful sequences [2]. That evolution shows why researchers and operators often layer an emulator or sandbox after static validation: emulation can exercise temporal behaviors, device state transitions, and interactions that are difficult to capture exhaustively as static rules.

Empirical and prototype evidence. Prototype systems that integrate generation with verification and execution-harnesses report heterogeneous gains: in some task regimes a verifier alone catches most dangerous errors, while in others emulation uncovers subtle runtime failures or environment-specific side effects that static rules miss [1], [3]. The practical implication drawn across this literature is that the marginal utility of a guarded (validator+emulation) pipeline depends on (a) validator coverage relative to the task distribution, (b) the fidelity of the emulator, and (c) the distributional properties of the input intents (e.g., constrained, parser-validated intents versus noisy natural-language requests). These prior empirical and system-level observations motivated a paired experimental design: by reusing the identical initial generated candidate across conditions (shared_initial_candidate semantics) we isolate only downstream guarded-stage effects (emulation observations or permitted repair invocations) rather than conflating them with generation variability.

Gaps and evaluation needs. While surveys and system reports argue for generate→validate→repair and for emulator-assisted validation as best practice for high-assurance NetOps, they do not provide large-scale, preregistered paired estimates of the marginal detection benefit under controlled generation semantics. The literature also documents important open questions that our study was designed to probe: how much additional coverage does sandboxed emulation provide when a strong deterministic validator already targets the same workflow patterns, and which task regimes (adversarial inputs, cross-vendor CLI heterogeneity, or stateful multi-step procedures) are most likely to reveal nontrivial guarded-stage benefits? By situating our paired confirmatory design amid these prior findings, we provide an artifact-grounded quantitative datapoint addressing one narrow slice of the broader evaluation agenda advocated in [1]–[3].

III. METHODOLOGY

Preregistration and experimental design. We preregistered a confirmatory paired-binary design (study_id f2c3d8b1-7a6e-4a9a-b2c3-5f8e1d2b6c7e). Primary estimand: paired_success_rate_difference_guarded_minus_baseline aggregated across confirmatory samples. Planned confirmatory sample: Npairs = 720 (planned episodes = 1440). Minimum effect-of-interest for power planning = 0.20 (20 percentage points). The preregistration specified inclusion/exclusion rules, missingness handling, multiplicity handling for exploratory secondary tests, and deterministic contamination controls.

Task bank, vendors, and inclusion. Confirmatory items were synthetic intent→configuration tasks from intent_configuration_repair_v1 covering three abstracted vendor-CLI families and three topology templates (leaf-spine, 3-node service chain, triangle). Inclusion required vendor-specific parser acceptance; the preregistered missingness policy permitted up to two deterministic reseeds for parser failures; irrecoverable parser failures would be deterministically excluded and replaced. In execution there were zero irrecoverable parser exclusions: complete_pair_count = 720 (see analysis/missingness_summary.csv).

Model configuration and sampling semantics. Declared/requested model: "gpt-5-mini" (provider field: openai_responses). The archived model configuration records max_output_tokens = 2000 and temperature = null (execution/model_configuration.json, sha256 a40e96e212ab87da...). Important clarification (preregistered and enforced): temperature = null indicates the client did not set an explicit temperature parameter; null does not imply temperature=0 or provider-side determinism. We enforced shared_initial_candidate semantics: for each pair we obtained exactly one sampled initial candidate (shared) which both baseline and guarded pipelines analyzed. Therefore any between-condition difference can only arise from (i) guarded-stage emulation observations not encoded as a validator rule, or (ii) the allowed downstream repair-stage invocation(s) recorded in run accounting.

Validator and guarded pipeline implementation. Baseline: deterministic_netops_validator_v5 performing full intent-constraint validation (syntax, command ordering/workflow, required-field presence, and policy-level constraints) and returning binary detected/not-detected per the preregistered scoring rule full_intent_constraint_validation. Guarded: baseline validator followed by an automated sandboxed-emulation harness executing the candidate against an emulated instance of the declared topology and producing runtime observations. The repair policy permitted at most one repair-stage model call per task; run accounting records exactly one repair-stage call in the confirmatory execution.

Execution controls, provenance, and deterministic reconciliation. The adapter family used: hosted_netops_gvr_v1. Execution manifest and deterministic reconciliation artifacts record completed_episode_count = 1440, complete_pair_count = 720, failed_episode_count = 0, and provider-call audit information including model_calls_used = 721 and stage_counts: shared_initial_generation = 720; repair = 1. Provenance and contamination controls (deterministic hashing, artifact-version checks) were applied per the preregistration; no contamination flags occurred for the confirmatory set.

IV. RESULTS

Primary confirmatory counts and test statistics (artifact-grounded). Analysis artifacts (analysis/paired_contingency_table.csv and analysis/condition_summary.csv) report the contingency counts: n11 = 719 (detected by both), n10 = 1 (guarded-only), n01 = 0 (baseline-only), n00 = 0 (neither). Marginal detection rates: baseline = 719/720 = 0.9986111111111111; guarded = 720/720 = 1.0. Paired estimate (guarded - baseline) = 0.0013888888888888884. Exact McNemar two-sided test p = 1.0. Paired-bootstrap percentile 95% CI (10000 resamples, bootstrap_seed = 1729) = [0.0, 0.0041666666666666667]. All numeric values below are reproduced verbatim from analysis/results.json and analysis artifacts.

Table 1 — Primary confirmatory summary (artifact-grounded). - Npairs = 720 (complete paired episodes) - Baseline successes = 719; Baseline success rate = 0.9986111111111111 - Guarded successes = 720; Guarded success rate = 1.0 - Paired difference (guarded - baseline) = 0.0013888888888888884 - Exact McNemar two-sided p = 1.0 - 95% bootstrap percentile CI = [0.0, 0.0041666666666666667] (10000 resamples, seed = 1729)

Discordant-pair attribution and repair accounting. Provider-call accounting and deterministic reconciliation identify a single repair-stage invocation (stage_counts: repair = 1) and a unique n10 discordant count. Because shared_initial_candidate semantics were enforced, the permissible causes for that single guarded-only detection are: (i) a guarded-stage emulation observation not encoded as a validator violation, or (ii) the single recorded repair-stage invocation. The confirmatory execution bundle contains the raw per-episode records; the unique discordant record can be located by searching execution/raw_results.jsonl (input-results SHA-256

= df367ecc40a6f0eaf402129260e11d8450884a4ea5fdb94a602e0d2513aef0e8) for the single entry where `baseline_score` = 0 and `guarded_score` = 1. Deterministic reconciliation is archived as `analysis/deterministic_reconciliation.json` (sha256 49731a2e52e049ccf5b53c35ddac2351ad75e7e95504dc4a42bcb096d0dec3aa) and its `provider_call_audit` confirms `model_calls_used` = 721 and `repair` = 1. The `raw_results` record for the discordant entry contains the `pair_id` and the `validator_trace` and `repair_provider_trace` fields that link the guarded detection to the documented repair-stage call; these artifacts are part of the canonical execution bundle referenced by the execution manifest.

Representative per-episode diagnostics. Representative scoring traces archived under `execution/scoring/task-000001-*.json ... task-000006-*.json` include `validator_trace` objects with `normalized_configuration`, `validation_before/after`, `parsed_command_count`, `required_command_count`, `violation_count`, and `validator_version` = 5.0. Those representative artifacts illustrate that the vast majority of `shared_initial` candidates complied with validator rules (`violation_count` = 0) and that sandboxed emulation largely corroborated validator findings in this curated task bank.

V. DISCUSSION

Why was the guarded-stage aggregate benefit negligible? The archived evidence supports three artifact-grounded factors and suggests concrete boundary conditions where guarded stages remain important.

1) Validator ceiling effect (artifact-grounded). The deterministic validator covered the explicit syntactic and semantic constraints exercised by the synthetic tasks: most `shared_initial` candidates parsed successfully and produced validator traces with `violation_count` = 0. Analysis artifacts record `n11` = 719 and `n10` = 1, so the baseline validator already detected or accepted nearly all conformant configurations (`baseline_success_rate` $\approx$ 0.9986). With validator coverage at this level there is little statistical headroom for the guarded stage to contribute aggregate detections at scale. Representative `validator_trace` fields (`parsed_command_count` and `required_command_count` in `execution/scoring/*.json`) confirm that tasks were largely within the validator’s rule set.

2) Task curation and inclusion policy concentrate cases inside validator coverage. The confirmatory bank used parser-validated synthetic items that explicitly encoded workflow patterns (shutdown→configure→restore, make-before-break up-link switchover, paired MTU changes). The inclusion rule required vendor-parser acceptance before pairing, and the missingness artifact shows zero irrecoverable parser failures for the confirmatory set (`analysis/missingness_summary.csv`). This curation intentionally constrained the evaluation to cases where the validator’s deterministic rules apply; it therefore reduces the prevalence of subtle runtime-only failure modes (race conditions, unexpected protocol interactions, or stateful side-effects) that emulation would be uniquely positioned to reveal.

3) `Shared_initial_candidate` semantics and observed repair activity. Per the preregistration we enforced `shared_initial_candidate`: the same single sampled candidate was analyzed by both conditions for each pair. `Provider-call` accounting and `deterministic_reconciliation.json` show `model_calls_used` = 721 (720 shared initial + 1 repair-stage call) and `stage_counts` `repair` = 1. Under these semantics, between-condition discordance cannot arise from different model outputs; it can only arise from (a) additional runtime observations surfaced by emulation that are not encoded as validator rules, or (b) permitted downstream repair actions. The single `n10` discordant count is explicitly mapped in the archived `raw_results.jsonl` record to the recorded repair-stage invocation, so the observed discordance is attributable to downstream guarded-stage activity rather than generation variability.

Practical interpretation and policy implications. The artifact-grounded conclusion is conditional: when a high-coverage static validator exists and inputs are constrained and parser-validated, automated sandboxed emulation may add negligible marginal detection at scale. Operationally, this suggests a cost/benefit trade-off: expensive emulation and canary infrastructure will have the most value when (i) validator coverage is known to be incomplete for target workflows, (ii) inputs include unconstrained or noisy natural-language intents, (iii) stateful multi-step changes introduce emergent runtime behaviors, or (iv) adversarial or fuzzed inputs are expected. In those regimes emulation can expose violations of invariants not expressible in the validator’s static rule set (for example, transient traffic blackholing under failure-induced routing changes or vendor-specific CLI side-effects). The archived per-episode traces (`execution/scoring/*.json`) provide concrete examples and can be used to build such adversarial testbeds.

Diagnostics useful to operators and evaluators. The run artifacts enable targeted diagnostics rather than blanket conclusions. For example: - Coverage auditing: compare `validator_rule_inventory` to `parsed_command_count` statistics across tasks to locate missing rules; `analysis/condition_summary.csv` and `execution/scoring/*` contain the necessary fields. - Discordant attribution: locate the unique discordant `raw_results.jsonl` entry (`baseline_score` = 0 and `guarded_score` = 1) and follow its `validator_trace` and `repair_provider_trace` to see whether the guarded detection arose from an emulation observation or an applied repair; `deterministic_reconciliation.json` confirms the single repair-stage call linkage. - Emulation fidelity checks: examine the emulation observation summaries in `validator_trace.validation_after` and compare them to `expected_final_state` in `task_manifest` entries to identify classes of runtime behaviors the emulator notices but the validator cannot express. These diagnostics are artifact-grounded and reproducible from the archived bundle referenced in the execution manifest.

Threats to validity and scope (artifact-grounded). Internal: `shared_initial_candidate` reduces variation in generation and isolates guarded-stage effects but constrains external generality — in practice operators may allow multiple candidate gener-

ations or stochastic sampling to improve correctness, which could change the relative benefit of guarded stages. Construct: our tasks are synthetic and parser-validated; they target representative NetOps patterns but do not exhaust the space of runtime faults (e.g., vendor implementation bugs, timing-sensitive state races). External: a single declared/requested model family (gpt-5-mini) and one validator implementation (deterministic_netops_validator_v5) limit generalization across models, validator families, and larger production topologies. Conclusion: power planning targeted a 0.20 minimum detectable paired difference; the observed paired difference (~ 0.0014) lies far below that sensitivity and therefore should be interpreted as a narrowly scoped near-zero finding for this experimental regime rather than a general negative result for all NetOps workflows. All of these limitations are documented in the preregistration and in analysis artifacts (missingness and reconciliation files).

Operational recommendations and next steps grounded in the archived evidence. To meaningfully evaluate guarded-stage benefits where they matter operationally, future controlled studies (using the same artifact-oriented framework) should: (1) include adversarially generated tasks and fuzzed intents designed to evade validator rules; (2) broaden vendor-CLI families and emulate vendor-specific OS behaviors with higher-fidelity toolchains; (3) allow multiple sampled initial candidates per task to measure interplay between generation variability and guarded-stage detection; (4) instrument canary/rollback-aware scoring that penalizes dangerous but syntactically valid changes; and (5) measure cost and latency trade-offs for emulation at scale. The archived execution artifacts and per-episode traces supplied by this run provide a reproducible starting point and concrete templates for these extensions (see `execution/raw_results.jsonl` and `analysis/deterministic_reconciliation.json`).

Summary. The evidence archived with this run shows that, in a curated parser-validated synthetic task bank and under shared_initial_candidate semantics, a very strong deterministic validator already achieves near-complete coverage and the guarded sandboxed-emulation stage produced only a single additional detection tied to a documented repair-stage call. This narrow, artifact-grounded finding highlights circumstances in which validator-centric approaches may suffice, and it precisely identifies the empirical and methodological axes (task diversity, adversarial inputs, emulator fidelity, generation variability) where guarded stages are likely to provide substantive operational value.

VI. LIMITATIONS

- Synthetic task bank and deterministic task generator produce limited ecological diversity and create a validator-ceiling effect; results may not generalize to noisier, adversarial, or production distributions.
- Single declared/requested model family (gpt-5-mini) and single validator implementation (deterministic_netops_validator_v5) limit generalizability across models and validator families.
- Preregistered shared_initial_candidate semantics (one initial generation reused across paired conditions) isolate guarded-stage contribution but remove between-condition generation variability; this design choice constrains discordance sources to downstream guarded-stage observations or repair calls.
- Sandboxed-emulation fidelity relative to vendor/OS production behaviors and large-scale stateful interactions is limited by the emulation harness used; higher-fidelity emulators could change outcomes in other regimes.
- Study power planning targeted a minimum detectable paired difference of 0.20; the observed aggregate effect (~ 0.0014) lies far below that design sensitivity and should be interpreted as a narrowly scoped near-zero finding for this experimental regime rather than a general negative result for all NetOps workflows.
- Provider-side nondeterminism and hidden caching remain residual uncertainties despite deterministic reconciliation procedures; these are documented in `analysis/deterministic_reconciliation.json` and `provider_call_audit` artifacts.

VII. CONCLUSION

We preregistered and executed a paired confirmatory study comparing a strong deterministic validator with the same validator augmented by sandboxed emulation on a curated synthetic intent→configuration task bank. Over 720 paired tasks the guarded pipeline produced a negligible aggregate detection gain (0.0013888888888888884) with exact McNemar $p = 1.0$ and 95% bootstrap CI [0.0, 0.004166666666666667]. Run accounting shows one repair-stage invocation corresponding to the single guarded-only detection; because shared_initial_candidate semantics were enforced, archived per-episode traces permit independent verification of that mapping via `execution/raw_results.jsonl` and `analysis/deterministic_reconciliation.json`. We interpret the result as arising from validator ceiling effects and curated task design; follow-up work should focus on adversarial and production-like task banks, multi-validator ensembles, and higher-fidelity emulation to identify regimes where guarded stages add substantive operational value.

Reproducibility pointers (compact). The canonical execution bundle archived by the run contains: `execution/execution_manifest.json` (execution summary and preregistration SHA-256), `execution/raw_results.jsonl` (input-results SHA-256 df367ecc40a6f0eaf402129260e11d8450884a4ea5fdb94a602e0d2513aef0e8), `analysis/deterministic_reconciliation.json` (sha256 49731a2e52e049ccf5b53c35ddac2351ad75e7e95504dc4a42bcb096d0dec3aa), `execution/model_configuration.json` (sha256 a40e96e212ab87da251ac0d6dbb75bc543afa13438b5a6b9ebd073597ffdd820), `analysis/paired_contingency_table.csv` (sha256 efe6e8e7016fa843a8226e911dbde829e943903d719101da8d956b3ff899133c), and `analysis/condition_summary.csv` (sha256 394bc690671b731194f9b42b4dcbe28fded6e2ec8cf898b79831225343c22c28).

The discordant episode is the unique `raw_results.jsonl` entry with `baseline_score = 0` and `guarded_score = 1`; that record contains the `pair_id` and validator/emulation traces that map it to the single repair-stage invocation recorded in `deterministic_reconciliation.json`. These artifacts are archived as the canonical reproducibility bundle referenced by the execution manifest and the preregistration record. (See Disclosure for exact manifest references.)

DISCLOSURE STATEMENT

Disclosure: Autonomous post-lock research run executed under the frozen master prompt supplied to the system. Declared/requested model: "gpt-5-mini" (provider recorded as `openai_responses`). Deterministic validator: `deterministic_netops_validator_v5`. Orchestration adapter family: `hosted_netops_gvr_v1`. Preregistration id: `f2c3d8b1-7a6e-4a9a-b2c3-5f8e1d2b6c7e` (preregistration SHA-256: `0169919fb8c5aedb2c5a78e031f1ab5c3aeda405cde1ae80fb2f864768595c1b`; recorded in the execution manifest). Initial master-prompt immutable reference: `provenance/master_prompt.txt`, SHA-256 = `1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582b b39b15ba`. Execution summary (artifact-grounded): `completed_episode_count = 1440`, `complete_pair_count = 720`, `model_calls_used = 721` (720 shared initial + 1 repair-stage call); `stage_counts` and provider-call audit are archived in `analysis/deterministic_reconciliation.json` (sha256 `49731a2e52e0...`). Human role and autonomy boundary: a human Research Director selected the broad NetOps topic family and constrained the pre-lock master prompt; after the autonomy lock the agentic pipeline autonomously performed literature selection, preregistration, code and prompt generation, experiment execution, analysis, peer review, and manuscript drafting. No manual human editing of technical manuscript content occurred after the autonomy lock. The execution manifest and archived artifacts named above serve as the canonical reproducibility bundle.

REFERENCES

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